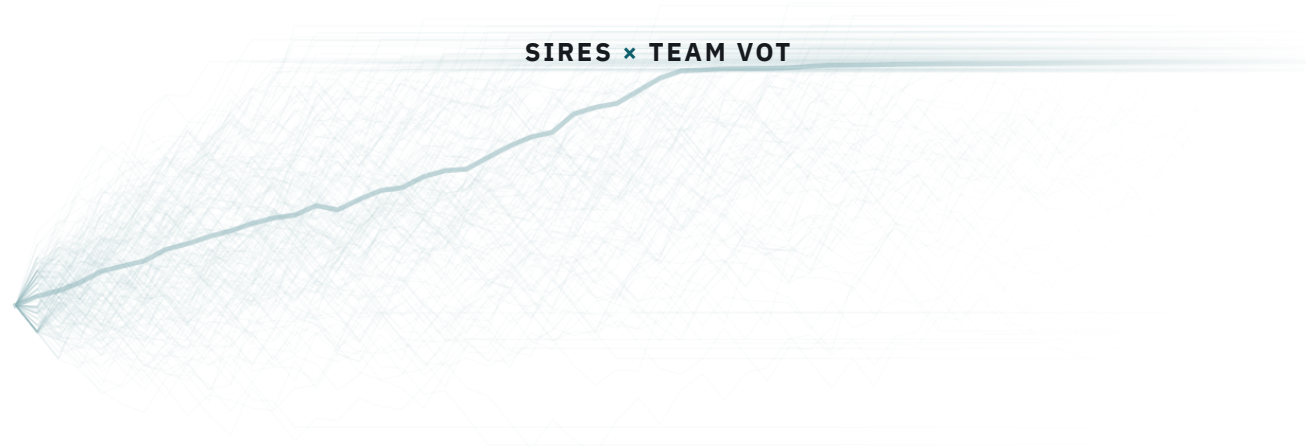




The Refill Effect

Aggressive orders, level memory, and a tradeable feature set in Nasdaq futures, measured, stress-tested, and published with its failures attached.



NQ / MNQ

CME

199M TICKS

235 SESSIONS

41,152 EVENTS

JULY 2026

Contents

What this paper shows	the three results, in one page	03
Introduction	the claim, and why we tested it like academics	04
The refill concept	zones, touches, and the one question that matters	05
The data and the feature set	199M ticks and the honest baseline	08
Selection	how grading the touches flips the sign	09
Where the money lives	the discovery that names the paper	10
The out-of-sample record	79 untouched sessions, trade by trade	12
An independent engine agrees	the same signals on software we didn't write	14
Why this is not overfitting	seven attacks on our own result	16
Prop-firm economics	pass rates, sizing, and the daily stop	19
The stat sheet	every number on one page	23

READ THIS FIRST

The language, in plain English

Zone: a price area where a burst of large aggressive orders traded. Someone with size fought there.

Feature: one measurable fact about a touch, known *before* it resolves (e.g. "did this zone hold earlier?"). The **feature set** is that checklist.

Expectancy: the average R per trade. Positive means the approach makes money over many trades.

Out-of-sample: data the model never saw during tuning. The only performance that counts.

AUC: grading skill from 0.5 (coin flip) to 1.0 (perfect). 0.63 = a real, modest, usable signal.

Touch: price returning to a zone later. Each touch is one event in this study.

R: result measured in risk units. Risk \$80, make \$240 → +3R. Lose the stop → -1R.

Profit factor: gross wins ÷ gross losses. Above 1.0 is profitable; 1.8 means \$1.80 won per \$1 lost.

Placebo test: rerun everything on scrambled labels. If the "edge" survives, your machinery is lying to you.

Drawdown: the give-back from a running peak. Prop evaluations fail you on it; we measure it everywhere.



IN ONE PAGE

What this paper shows

We measured one idea on 199 million ticks of Nasdaq futures data: price levels built by **large aggressive orders**, and the **memory** of whether those levels were defended when price returned. Three results follow.

01 / THE RAW CONCEPT LOSES

Fading every touch of every level loses **-0.285R per trade** after costs. Only 42% of touches hold. Anyone selling "just fade the level" is selling this line.

02 / SELECTION FLIPS THE SIGN

A small statistical model reads each touch's context: has this level held before, was it built by real size, where does it sit in the day, and it separates the touches that hold from the ones that break (**AUC 0.63** against a 0.51 placebo). Taking only the best-graded touches flips expectancy to **+0.14R per trade**, on data the model never saw.

03 / THE EDGE IS THE REFILL

The profit concentrates almost entirely in one execution style: **resting passively inside the level, where the defenders reload**. The same signals traded with market orders lose money.

We then attack our own result with placebos, reversed splits, rotating folds and a 100-configuration audit, and **we publish the losing quarter along with the winning ones.**

Companion reading: *Origin of the Move*, the education paper for this feature set, free at kanji.org.uk.



01 Every trader has seen it

Price falls into a level. Heavy selling hits the bid. Nothing happens. The offers absorb, the level refuses to break, and price leaves. When price comes back to that level later, it holds again.

Traders call the mechanism a **refill**: the passive orders defending the level are pulled, filled, and quietly replaced. The level "remembers" who defended it, because those participants are still there.

The claim is easy to say and easy to sell. It is much harder to prove. Harvey and Liu [2014] showed that most published trading strategies are false discoveries once you account for how many things were tried. That warning applies to us as much as to anyone.

So this paper does something unusual for a trading education group: it puts the concept through the same machinery an academic would use, and publishes the results either way. We audited every input for hindsight, tested against scrambled-label placebos, reversed the data split, rotated the folds, and re-ran everything on an independent backtest engine we do not control.

What we found, in one sentence: the refill concept survives, but not where most people look for it.

The entry signal alone is modest. The **selection layer** (which levels have memory) and the **execution layer** (being the refill, rather than chasing it) carry nearly all of the money.

We also report what did **not** survive: raw aggression with no context, tight scalp brackets that lived on fill assumptions, and a losing final quarter. All of it is in here, because that is what makes the rest believable.



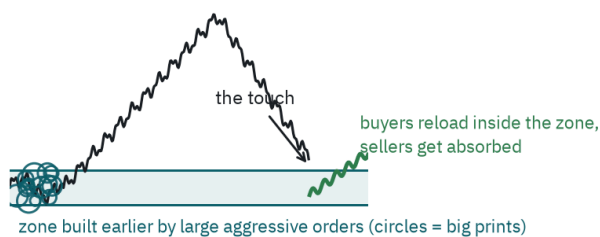
The refill concept

When a burst of large aggressive market orders trades at one price area (sixty, eighty, a hundred contracts hitting in seconds) that area stops being a random price. Somebody with size chose to fight there. We call the cluster of prints a **zone**.

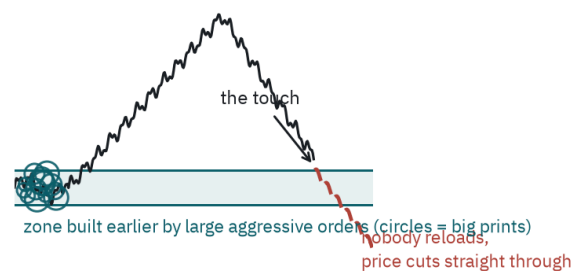
The next time price returns to the zone, exactly one question matters: **are the defenders still there?** If they are, the touch is absorbed, the zone refills, and the level **holds**. If they are gone, the zone is just ink on a chart, and price trades straight **through** it.

The two outcomes of a touch

The level **HOLDS**: defenders are still there



The level **BREAKS**: defenders are gone



A schematic, not a trade example. A zone is built by clustered large aggressive prints (circles, sized by print size). Price leaves, then returns: that is the touch. Everything in this paper exists to answer, *before the touch resolves*, which of these two pictures you are in.

A textbook example, on the real chart



A real MNQ session in Quant Charts, the platform every statistic here was validated on. Red sell-bubbles hammer a level, green buy-bubbles absorb them, and price turns and trends up. That absorption is the refill.



Test it before you trust it

IT IS MEASURABLE BEFORE THE FACT

Everything that defines the zone (print sizes, width, location in the session's structure, and whether it held an **earlier** touch) is known before the next touch happens. Nothing peeks at the outcome. That makes it a legitimate forecasting problem, not a story told in hindsight.

IT HAS AN ECONOMIC REASON TO EXIST

A defended level is not a chart pattern; it is **inventory**. Participants who accumulated there defend the position because letting the market prove them wrong is expensive. Edges with a mechanism have a reason to persist. Edges without one are usually noise that got lucky.

Testing it honestly means turning the idea into numbers. Every zone becomes a row of data. Every return of price to a zone, every **touch**, becomes an event with a known outcome: the level held, or it broke. And every fact we could have known *before* that outcome becomes a **feature**: a checklist item a trader could read off the chart in real time.

That checklist, the **feature set**, is what this paper is really about. Not an indicator, not a magic entry: a set of observable facts about a level that, together, tell you whether the defenders are likely still there.

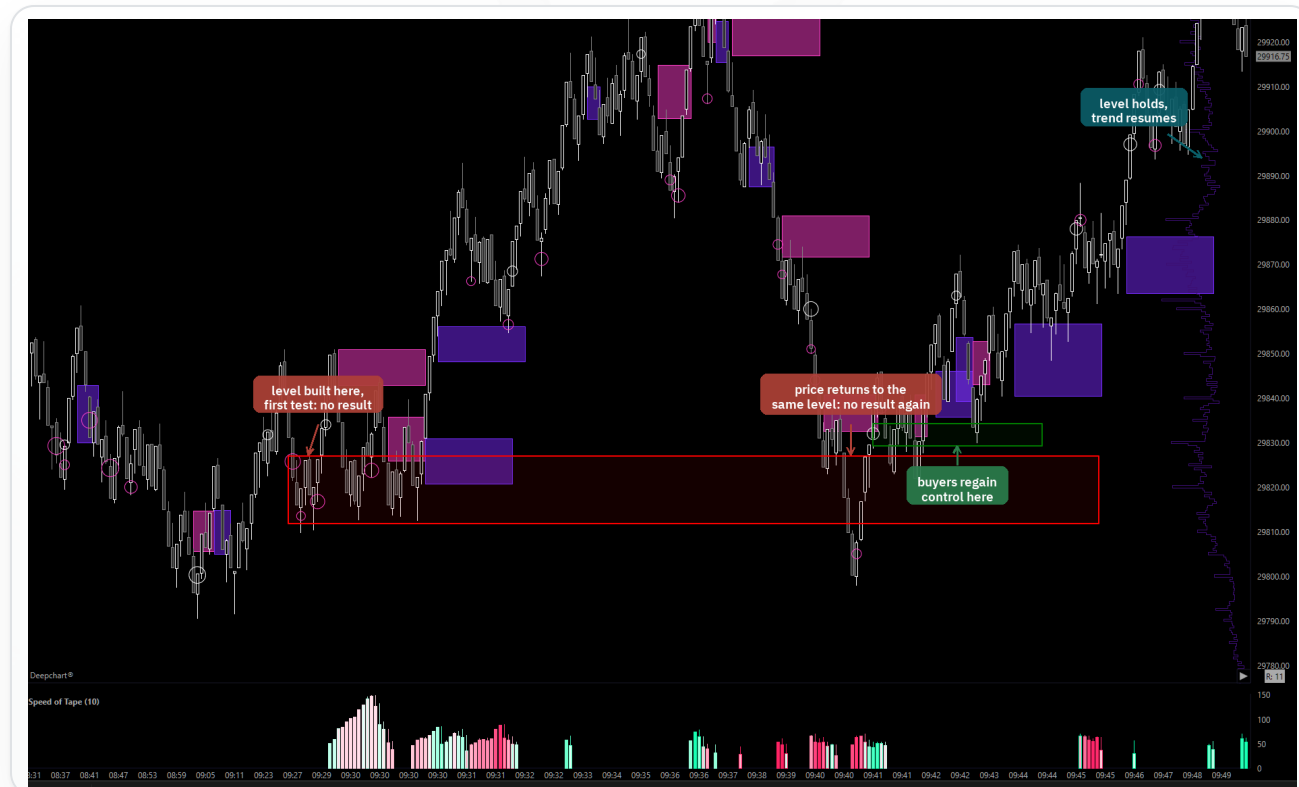
The next sections measure it end to end: the data and the baseline, where the profit actually lives, the out-of-sample record, the overfitting battery, and what it means inside a funded-account evaluation.



What the aggressive orders look like

Before any modelling, this is the raw material, read live off the DOM the way it actually happens. The rectangles below are zones built by clusters of large aggressive orders; the circles are individual large prints. The red and green boxes are the read in real time: a level gets tested, gives the aggressor no result, and either holds again or finally gives way.

A live read: the level gets tested twice before buyers take it



A real NQ session. The level (red) is built early and tested twice, once on the first touch and again on a sharp sell-off back into it, and both times the sellers get nothing: price holds, not breaks. Right after the second test, buyers build a small base (green) just above the level and take control. That is the read this whole paper tries to quantify: not the entry trigger, but whether the defenders are still there when price comes back.



03

The data and the feature set

199M

TICKS, AGGRESSOR-
SIDE

235

REGULAR SESSIONS

41,152

ZONE-TOUCH EVENTS

Dec 24–

Nov 25

CME NQ / MNQ

Each touch carries roughly twenty pre-touch features, in four families, every one of them a fact a trader could check on the chart before the trade:

FAMILY	THE QUESTION IT ANSWERS	EXAMPLES
Memory	Has this zone been defended before?	held earlier this session; prior-day defence
Construction	Was it built by real size?	number & size of aggressive prints; zone width
Location	Is it at a real auction edge?	value-area edges, session extremes, distance from open
Flow & state	What is happening as price arrives?	delta into the touch, approach speed, balance vs imbalance

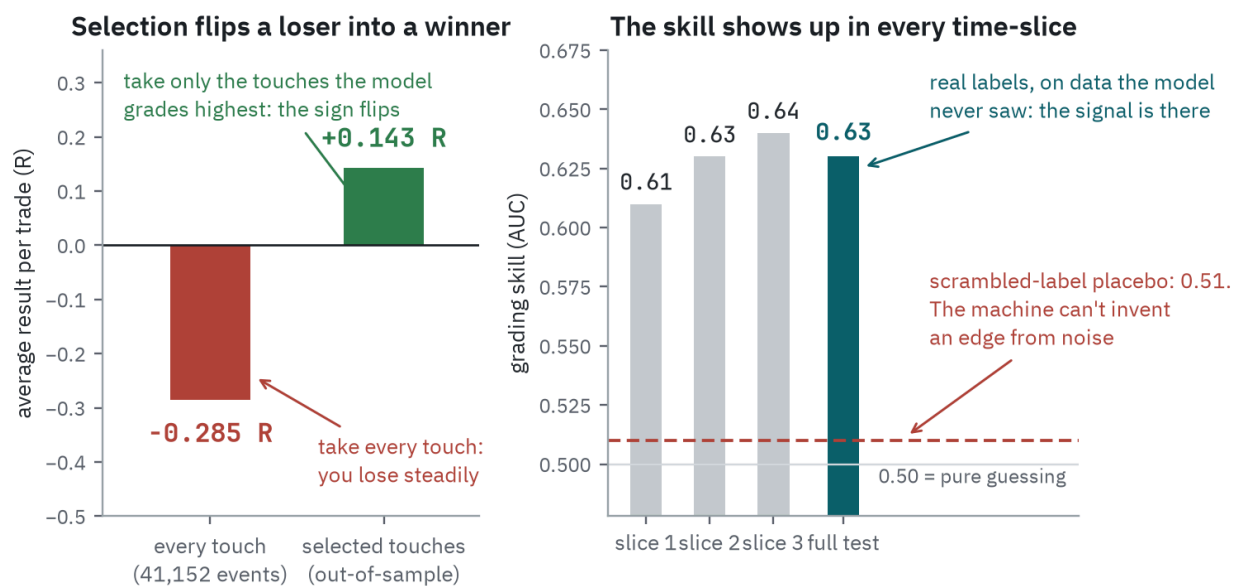
Start with the honest baseline. Only 42% of touches hold. Fading every touch (buying every support touch, selling every resistance touch, no questions asked) loses **-0.285R per trade** after costs. The raw concept, traded indiscriminately, is a losing strategy. That is the starting point, not the punchline.



Selection is what changes the sign

A deliberately simple statistical model reads the feature checklist for each touch and produces one number: how likely is this level to hold? It trains only on the past and grades only on the future, never the other way around. Out-of-sample, the touches it ranks highest hold 63% of the time; the ones it ranks lowest hold 25%.

Trading everything loses; selecting on memory wins



Left: the average result per trade, taking every touch versus only the model-selected touches, both measured after costs.

Right: the model's grading skill in three separate time-slices it never saw, versus the same machinery trained on scrambled labels.

ONE FINDING CUTS AGAINST OUR OWN BRANDING

When we decompose the model, the **memory and location** families do almost all the work; raw order-flow features alone barely beat a coin flip (AUC 0.54). Aggression *builds* the level, but it is the level's memory that predicts the next touch. We would rather report that than sell a prettier story.



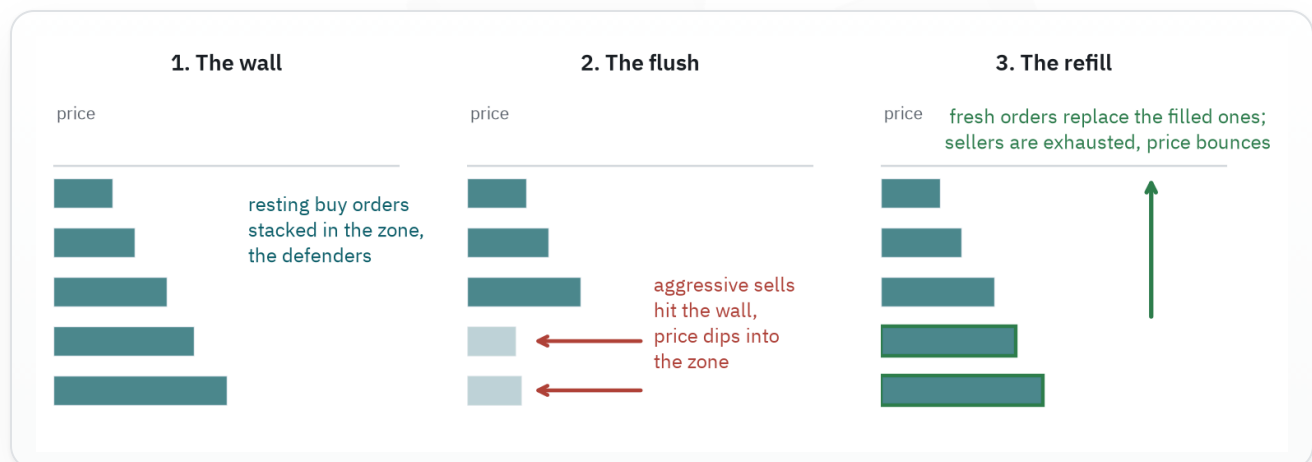
04 Where the money actually lives

This is the section that gives the paper its name. It began as a failure.

Our first backtests used a tight scalp bracket and looked wonderful. Then we replayed the identical signals against real tick-by-tick data and found the result depended entirely on which side of the bracket was assumed to fill first inside each bar. Optimistic assumption: heavily profitable. Pessimistic: heavily losing. Honest middle: a coin flip. The "edge" was an artifact of fill assumptions, so we threw it away and started over at the tick level.

The tick data then explained **why**, and the reason is the refill itself: around a genuine defended level, the median eventual winner first dips **18 ticks past the touch** before it works. The defence does not happen at the front edge of the zone. It happens **inside** it, where the resting orders reload.

The refill, one layer deeper: what happens in the order book



Each bar is resting buy orders at one price. The wall gets hit, price dips inside the zone, and if the defenders are real, fresh orders replace the filled ones before the sellers can break through. The trade is to be one of those replacing orders.

Same signal. **Opposite side of the mechanism.**



Identical signals, opposite outcomes

MARKET ORDER AT THE TOUCH

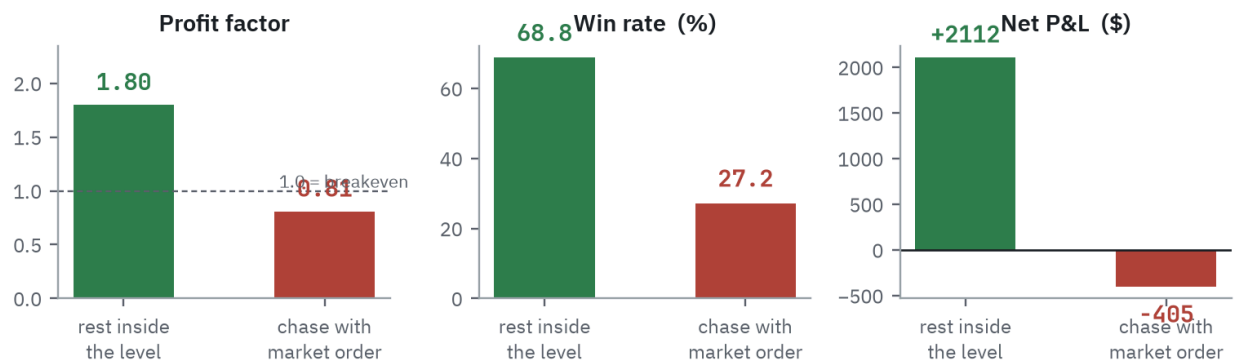
You do what the losing aggressors do: pay the spread to push into a wall, enter before the flush, and eat the full 18-tick dip against a tight stop. **You pay for the defence.**

RESTING LIMIT INSIDE THE ZONE

You are filled by that same flush, at the price where the refill is defending, alongside the defenders. **You are the refill**, and you collect the premium the impatient side pays.

The edge is the refill

Everything identical: signals, quarter, engine, costs. Only the ENTRY TYPE changes.



Same selected signals, same quarter (Q1), same independent tick engine. The resting-limit entries fill on 64 trades; the market-order entries fill on 312. The limit fills less often precisely because it demands the flush that defines the setup.

Nothing changes between the two columns except the entry order type: not the signals, not the period, not the engine, not the costs. Resting inside the zone earns a profit factor of **1.80**; chasing the touch with market orders earns **0.81** and loses money.

This is also why the effect is hard to data-mine into a lie: it is not a chart pattern, it is a **mechanism with a fee attached**, and the fee changes sign depending on which side of the order book you stand on.

This execution, resting inside the level rather than chasing it, is what Ethos trades live every session. Seven day free trial: whop.com/ethos-orderflow



05 The out-of-sample record

The deployed configuration in one sentence: at a model-selected zone, rest a limit order **12 ticks inside the level**, stop 32 ticks, target 96 ticks, cancel after 30 minutes, one position at a time, with a 1-tick round-trip cost and 1 tick of stop slippage charged.

All tuning used the first 156 sessions. The final **79 sessions were never touched** until the configuration was frozen:

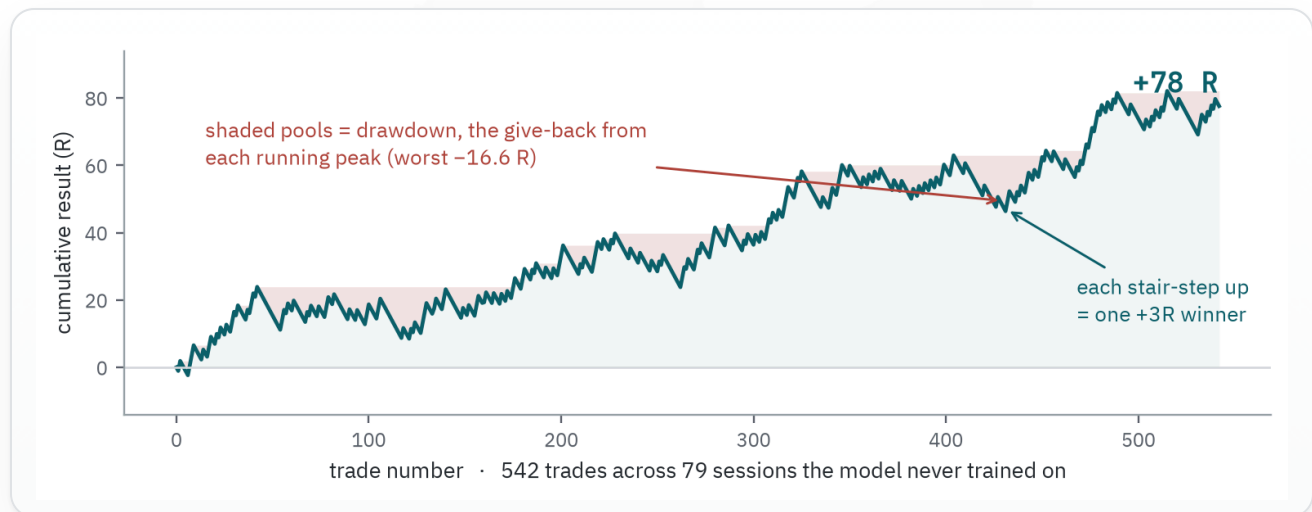
+0.143RPER TRADE, AFTER
COSTS**1.19**

PROFIT FACTOR

+78RCUMULATIVE, 542
TRADES**~6.8**

TRADES PER DAY

Out-of-sample equity, tick-resolved, after costs



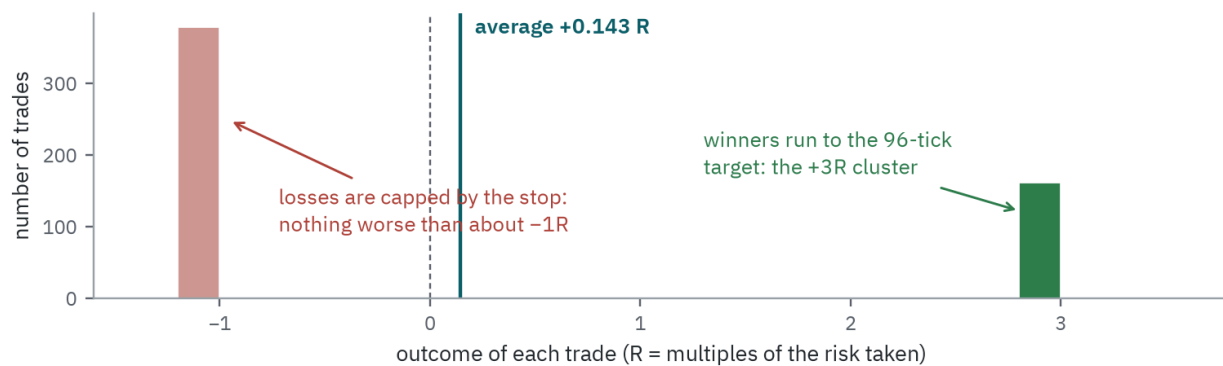
Every one of the 542 trades across the 79 held-out sessions, in sequence. No parameter was chosen using any of this data.



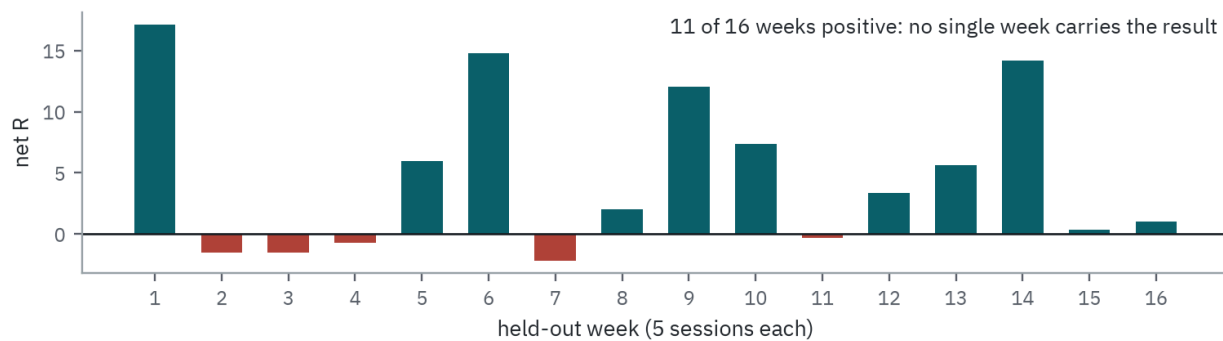
The shape of the returns

Averages hide character. The distribution below is what a 30%-win, 3-to-1 strategy actually feels like to trade: many small bounded losses, a steady cluster of full winners, and nothing worse than the stop.

Every trade outcome, out-of-sample



Week-by-week consistency



Net R for each block of five held-out sessions.



An independent engine agrees

Equity curves from your own research code deserve suspicion. So we embedded the same signals into a strategy for **Quant Charts'** tick-level backtester, an engine we did not write, with its own fill logic, and ran it over the full year at its default sizing.

PERIOD	TRADES	WIN RATE	PROFIT FACTOR	NET P&L
Q1 Dec-Mar	64	68.8%	1.80	+\$2,112
Q2 Apr-Jun	60	55.0%	1.28	+\$928
Q3 Jul-Sep	65	55.4%	1.73	+\$1,410
Q4 Oct-Nov	34	52.9%	0.76	-\$647
Year	223	58.7%	1.45	+\$3,803

Per-trade expectancy on this engine $\approx +0.07R$, about half the research simulation's $+0.14R$, because its fills are stricter. We quote both rather than the friendlier one.

NOTE THE LOSING QUARTER

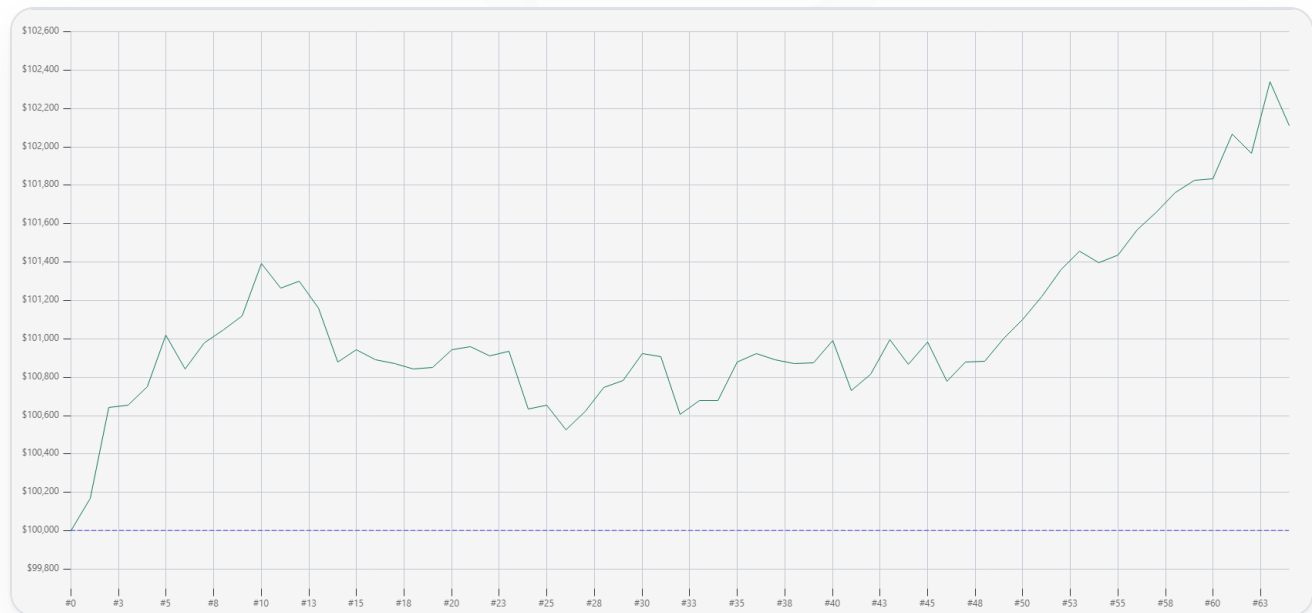
Q4, the most recent slice, lost money at **every** parameter setting we tried. The edge is regime-dependent: weakest when volatility compressed late in the sample. We flag it and count it in every statistic, because an edge that only appears when losing periods are trimmed away is not an edge.



Straight from the engine

Nothing below is redrawn or re-styled by us. These are the raw Quant Charts outputs for the first quarter, exactly as the software renders them. The equity curve is one contract, compounding from a \$100,000 base across the 64 trades the engine took.

Quant Charts equity curve, Q1, one contract



The engine's own performance summary

Aggression-Memory Tick · rest_min=30 variant=3 entry_in=12 t_minutes=30 sl_ticks=32 tp_ticks=96

Net P&L	+\$2112.39	Trades	64
Win Rate	68.8%	Profit Factor	1.80
Sharpe	0.22	Max DD	-\$1051.98
Avg Trade	+\$33.01	Largest Win	+\$475.01
Largest Loss	-\$300.62		

Net +\$2,112 across 64 trades, 68.8% win rate, profit factor 1.80, worst drawdown -\$1,052. Read straight off the Quant Charts results panel, the same figures quoted in the table on the previous page.



Why this is not overfitting

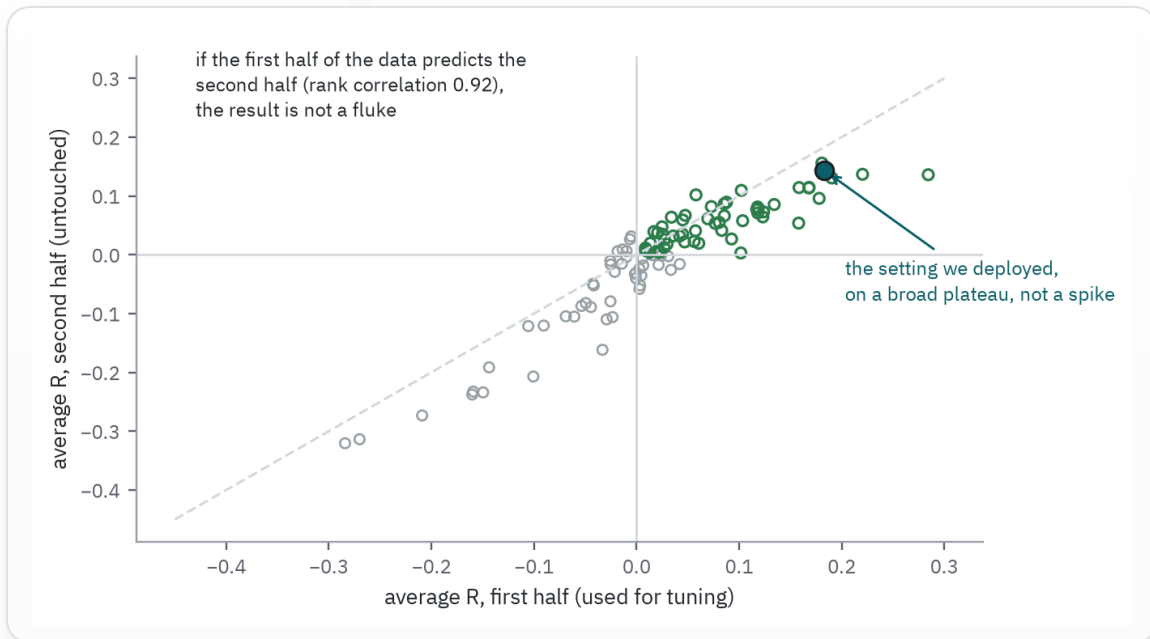
A profitable backtest is the cheapest object in finance. Anyone who mines enough combinations finds a curve that goes up. So we attacked our own result seven ways. It survives all seven. This is the strongest claim in the paper.

- 01 The hindsight audit.** Every feature was tested for information leaks. One failed: a stored field that predicted outcomes even at a zone's *first* touch, impossible without contamination. Removed and rebuilt cleanly. The edge stood without it.
- 02 The scrambled-label placebo.** Retrain the identical pipeline on shuffled labels: grading skill collapses to a coin flip, and only **0.2%** of shuffles match the real result. The machinery cannot manufacture an edge from noise.
- 03 Strict chronological split.** All choices were made on the first half of the data; only the untouched second half is quoted throughout this paper.
- 04 The reverse split.** Train on the *late* data, test on the *early* data. The edge holds, slightly stronger. A one-regime artifact fails exactly this test.
- 05 Rotating folds.** Three time-slices, each held out in turn: mean R **+0.14 to +0.22**, win rate 63–67%, grading skill 0.61–0.64 in every slice. No single lucky period carries the result.
- 06 The parameter plateau.** All 100 exit configurations: performance on the first half of the data predicts the second half with rank correlation **0.92**, and the whole neighbourhood of the deployed setting stays positive. The next page shows it.
- 07 A mechanism, not a pattern.** Overfit patterns do not care how you execute; a real liquidity mechanism does. The profit lives exactly where the mechanism says it must, on the passive side of the level, and dies when you chase.



A plateau, not a spike

One hundred configurations: the first half predicts the second

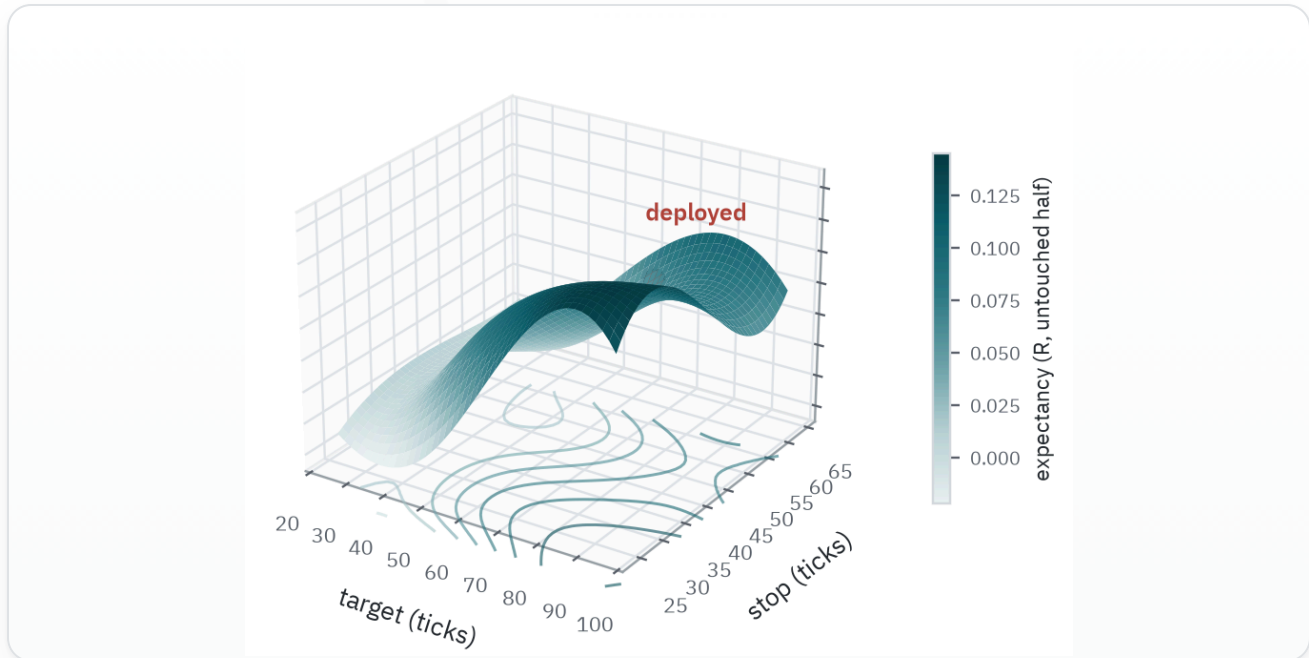


Each circle is one exit configuration (stop $\times$ target $\times$ entry depth), placed by its average R on the first half of the data (x) and the untouched second half (y). Green circles are profitable in both halves. If the result were curve-fit, this cloud would be shapeless.

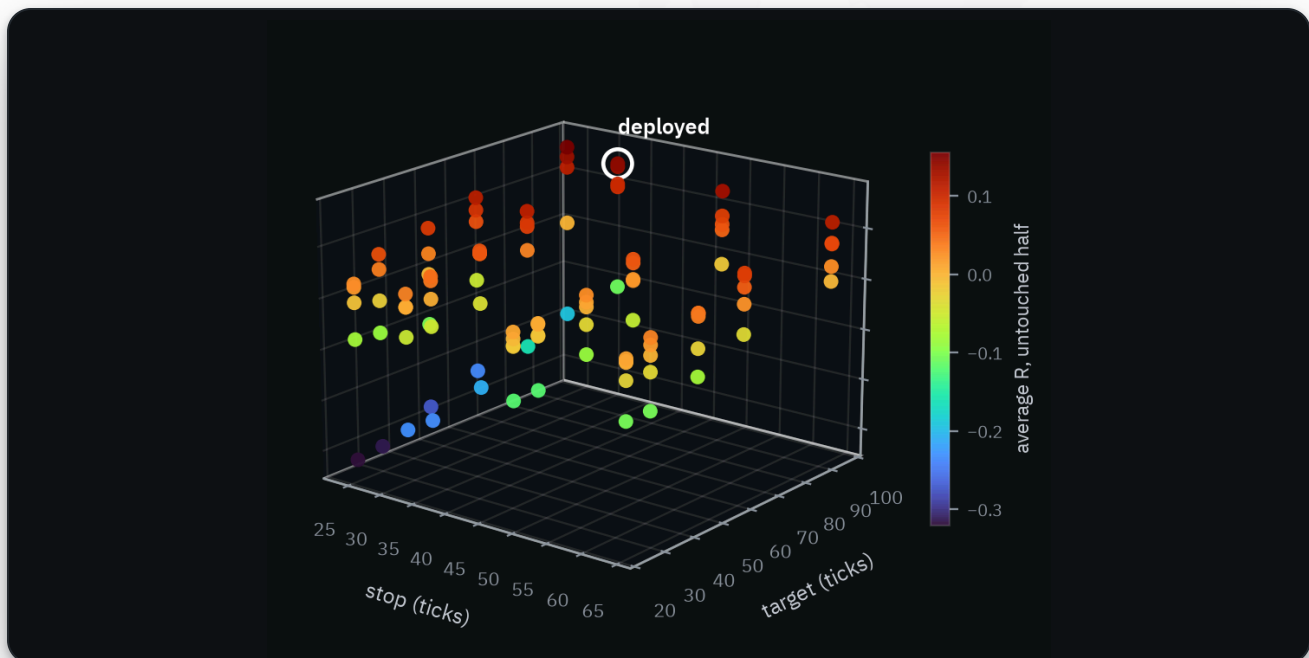


The whole parameter space, two ways

The ridge: expectancy across every stop and target



The same space, point by point



Both views show all 100 configurations on the untouched half of the data. A curve-fit edge is one bright dot in a sea of noise; this is a coherent warm region. The parameters barely matter, the concept does.

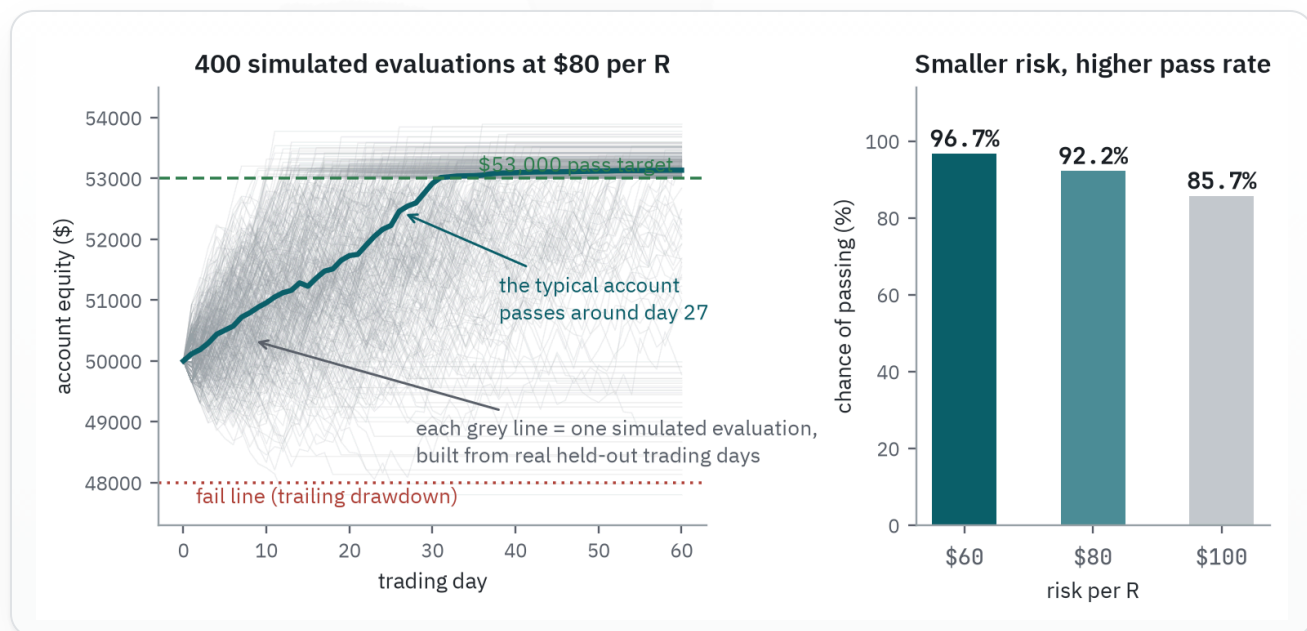


07 Inside a funded-account evaluation

A per-trade edge of $+0.14R$ is modest. What makes it valuable is its **shape**: many small bounded losses, steady drift, and a fat right tail from the 3-to-1 target. That shape is close to ideal for a prop evaluation, which is a race to $+\$3,000$ before a $\$2,000$ trailing drawdown, not a race for maximum profit.

We replayed the held-out daily results 4,000 times through a Topstep-style 50K evaluation, with the consistency rule enforced and one self-imposed policy: **stop trading any day that reaches $-4R$.**

The evaluation as a barrier problem



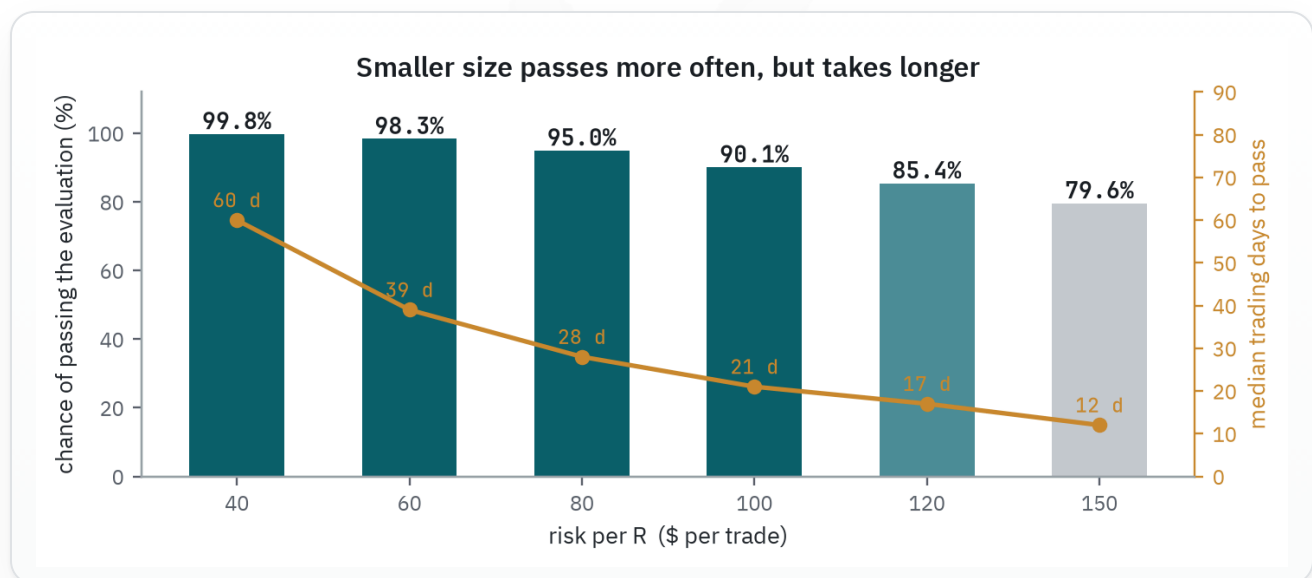
Read the left panel as a fan of possible futures. Almost every path clears the target; only the unluckiest sequences of losing days brush the fail line. The right panel is the headline: at $\$80$ of risk per R, the strategy clears a Topstep-style evaluation **92% of the time**. The rest of this chapter takes that apart: how size trades against speed, why the daily stop matters so much, and what a pass is actually worth.



Size sets the trade-off between odds and speed

Bet size is the one dial you fully control, and it works in two directions at once. Risk less per trade and the trailing drawdown almost never catches you, but it takes many more trading days to grind out the \$3,000 target. Risk more and you get there faster, at the cost of a few percent of pass rate. There is no free lunch, only a curve you get to choose a point on.

Pass rate and time-to-pass, by risk size



Teal bars: chance of passing (left axis). Amber line: median trading days to reach the target (right axis). At \$60 per R you pass ~97% of the time but wait ~7 weeks; at \$150 you pass in under two weeks, roughly four in five attempts.

PATIENT SIZING

\$60–80 per R: pass rate in the 90s, but you must be willing to trade for a month or more without forcing the pace.

FASTER SIZING

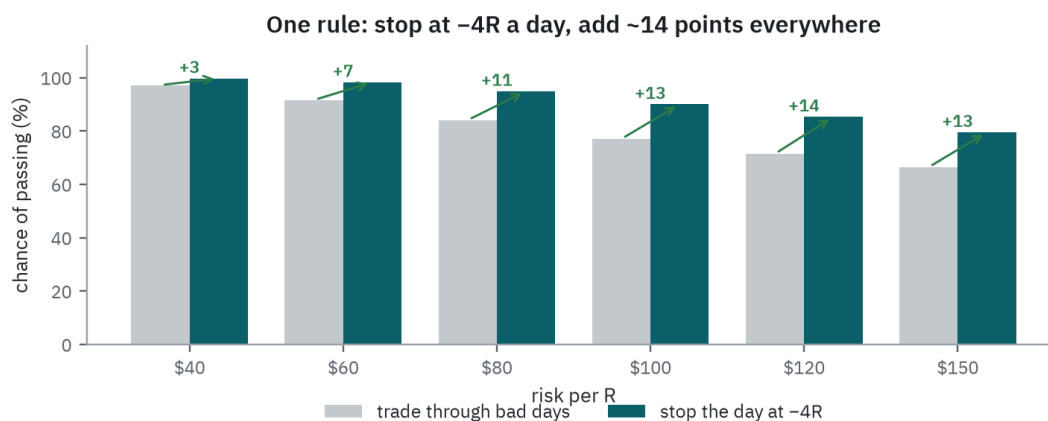
\$100–150 per R: a pass in 10–20 days, at 80–86% odds. Right when you have several attempts queued and want turnover.



The daily stop, and what a pass is worth

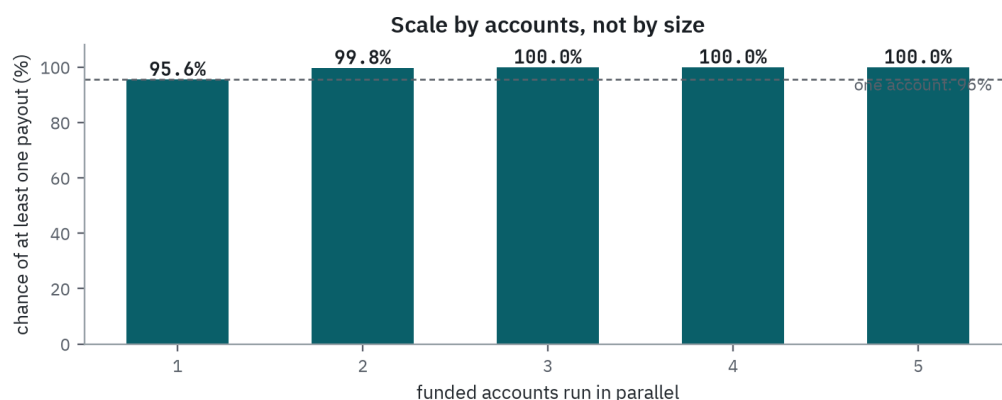
The single most powerful rule we found has nothing to do with the entry: **stop trading for the day once you are down four risk units**. A trailing-drawdown evaluation is asymmetric: one runaway losing day ends the account, while no single good day is worth the same in reverse. Cutting the tail of bad days off is worth about fourteen points of pass rate at every size we tested.

The -4R daily stop, measured



Passing is only half the economics. The payout on a funded account is **structurally capped** near \$2,000 per cycle by the same trailing drawdown, and no edge, however large, lifts that ceiling. The lever for total income is not size, it is **count**: run several funded accounts in parallel and the chance that at least one reaches a payout climbs fast.

Reach a payout: scale by accounts, not by size



A single funded account reaches a first payout ~96% of the time at \$80 per R; three in parallel make at least one payout a near-certainty. Many small, capped, high-probability payouts, not one big score.

The sizing and the daily-stop policy in this chapter are the same rules run inside Ethos. Seven day free trial:

whop.com/ethos-orderflow



08 Limitations, stated plainly

The out-of-sample window is 79 sessions, one regime slice, and the most recent quarter lost money. Limit fills assume a fill when price trades at or through the level; real queues are less kind. Costs are modelled, not lived.

The per-trade edge is thin enough that **discipline, not the signal**, decides whether it survives contact with a human. And every number here is historical simulation: it describes what was, not what will be. Nothing in this paper is financial advice, and no result guarantees any future outcome.

SOURCES

References

Harvey, C. R., and Y. Liu. "Evaluating Trading Strategies." *The Journal of Portfolio Management*, 40th Anniversary Issue (2014), pp. 108–118.

López de Prado, M. *Advances in Financial Machine Learning*. Wiley, 2018: meta-labelling; purged walk-forward cross-validation.

Steidlmayer, J. P., and S. B. Hawkins. *Steidlmayer on Markets: Trading with Market Profile*. Wiley, 2003: auction framework.



Complete statistics on one page

STATISTIC	VALUE	BASIS
DATA		
Instrument / period	NQ & MNQ, Dec 2024–Nov 2025	CME tick data, aggressor side
Ticks / sessions / events	199M / 235 / 41,152	all touches, no discretion
THE RAW CONCEPT		
Touches that hold	42%	base rate
Fade every touch, mean R	-0.285R	the honest baseline
SELECTION		
Touch-grading, out-of-sample	AUC 0.63	purged walk-forward
Scrambled-label placebo	AUC 0.51, 0.2% match	placebo control
Hold rate, worst → best decile	25% → 63%	monotonic
Dominant features	memory + location	flow alone: AUC 0.54
EXECUTION / THE REFILL EFFECT		
Median winner's dip past the touch	18 ticks	why chasing fails
Resting limit: win / PF / P&L	68.8% / 1.80 / +\$2,112	independent engine, Q1
Market order: win / PF / P&L	27.2% / 0.81 / -\$405	same signals, same engine
OUT-OF-SAMPLE RECORD / 79 UNTOUCHED SESSIONS		
Trades / per day	542 / ~6.8	limit 12t in, 32t stop, 96t target
Mean R / PF / win rate	+0.143R / 1.19 / 30%	tick-resolved, after costs
Cumulative / weekly consistency	+78R, 11 of 16 weeks green	no single week carries it
Independent engine, year	+\$3,803, 3 of 4 quarters	≈ +0.07R/trade, stricter fills
Worst quarter (most recent)	-\$647, PF 0.76	disclosed, included
OVERFITTING BATTERY		
Hindsight audit	1 leaky feature removed	edge stands without it
Reverse split / rotating folds	holds, +0.14 to +0.22R	all positive, AUC 0.61–0.64
Parameter grid, train → test	rank corr 0.92	100 configs, plateau
FUNDED-ACCOUNT EVALUATION / 4,000-PATH MONTE CARLO		
Pass at \$60 / \$80 / \$100 per R	96.7% / 92.2% / 85.7%	consistency + -4R daily stop
-4R daily stop	+14 pts pass rate	largest single lever
Reach first payout (\$80/R)	95.5%	payout ≈ capped \$2k/cycle
Three staggered accounts	P(≥1 payout/cycle) ≈ 99.6%	scale by accounts, not size

All figures are historical simulation after modelled costs. Educational statistics, not a promise of future performance.

To watch this feature set traded live, the trial is on the last page.





The statistics found the levels. Reading them live is **the craft**.

Every input in this paper, the zones, the memory grading, the resting entries, the minus-4R day, is taught and traded live inside Ethos Order Flow. Three live sessions a week on NQ order flow, the full archive, and the thesis work that supplies direction. The trial runs seven days with everything unlocked.

START THE 7 DAY FREE TRIAL

whop.com/ethos-orderflow

Two minutes to start. Cancel from your Whop settings in two clicks if the room is not for you.

You have read the losing quarter. Results inside are reported to the same standard.

Ethos Order Flow is an educational community. Nothing in this paper or inside Ethos is financial advice, a solicitation, or a promise of profit. Futures trading carries substantial risk of loss and is not suitable for every person. All statistics herein are historical simulations after modelled costs; past or simulated performance does not guarantee future results.

Companion reading: *Origin of the Move*, the education paper for this feature set, free at kanji.org.uk